

CSE 4631

Introduction to Signals & Systems

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What is a Signal?

Definition

A **signal** is any physical quantity that carries **information** and varies as a function of one or more independent variables such as time, space, or frequency.

Mathematically, a signal is a **function**:
 $s = f(\text{variable})$

Key properties:

- Carries information from a source to a destination
- Can be measured, recorded, and reproduced
- Can be processed to extract or enhance meaning

Real-World Examples

- **Temperature:** $T(t)$ — varies with time
- **Speech / Audio:** $s(t)$ — pressure wave
- **Image:** $I(x, y)$ — brightness at pixel
- **ECG:** $v(t)$ — heart electrical activity
- **Seismic:** $g(t)$ — ground vibration

Multichannel Signals

Definition:

- A signal may consist of multiple components from different sensors.
- Such signals are represented in **vector form**.

Example (Vector Signal):

$$\mathbf{s}(t) = \begin{bmatrix} s_1(t) \\ s_2(t) \\ s_3(t) \end{bmatrix}$$

- Each component represents a separate measurement.
- Example: Earthquake ground acceleration (3 sensors).
- Example: ECG signals (3-lead, 12-lead systems).

Real vs Complex Signals

Real-Valued Signal:

$$s(t) = A \sin(3\pi t)$$

- Takes only real values.

Complex-Valued Signal:

$$s(t) = Ae^{j\omega t}$$

- Contains real and imaginary parts.
- Widely used in communication systems.

One-Dimensional vs Multidimensional Signals

One-Dimensional Signal:

- Depends on a single variable (e.g., time)
- Example: $s(t)$

Multidimensional Signal:

- Depends on multiple variables
- Example: Image signal

$$I(x, y)$$

- x, y represent spatial coordinates

Examples of Multidimensional Signals

2D Signal (Image):

$$I(x, y)$$

- Brightness depends on position (x, y)

3D Signal (Video):

$$I(x, y, t)$$

- Brightness depends on space and time

Color Video Signal:

$$\mathbf{I}(x, y, t) = \begin{bmatrix} I_R(x, y, t) \\ I_G(x, y, t) \\ I_B(x, y, t) \end{bmatrix}$$

- Three channels: Red, Green, Blue

Key Takeaways

- Signals can be:
 - Scalar or vector
 - Real or complex
- Multichannel signals:
 - Multiple signals grouped together
- Multidimensional signals:
 - Depend on multiple variables (space, time)
- Example:
 - Image $\rightarrow$ 2D
 - Video $\rightarrow$ 3D
 - Color video $\rightarrow$ Multichannel + Multidimensional

Types of Signals

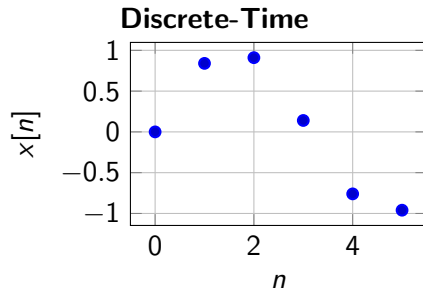
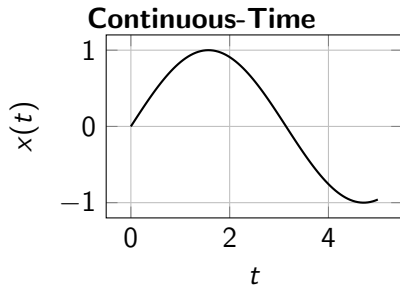
Continuous-Time (CT) Signals

- Defined for **every** instant $t \in \mathbb{R}$
- Also called **analog** signals
- Notation: $x(t)$
- *Examples:* Human voice, temperature sensor voltage, AM radio

Discrete-Time (DT) Signals

- Defined only at **integer** indices $n \in \mathbb{Z}$
- Obtained by **sampling** a CT signal
- Notation: $x[n]$
- *Examples:* Digital audio, stock prices, sampled sensor data

Continuous vs Discrete-Time Signals



Continuous-Valued vs Discrete-Valued Signals

Key Idea: Signal values can be either continuous or discrete.

Continuous-Valued Signal

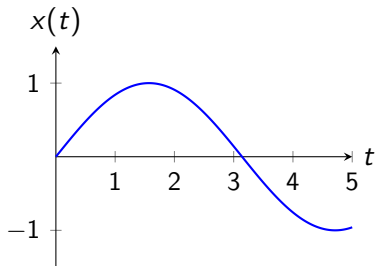
- Takes **any value** in a given range
- Infinite possible amplitudes
- Example: temperature, audio voltage

Discrete-Valued Signal

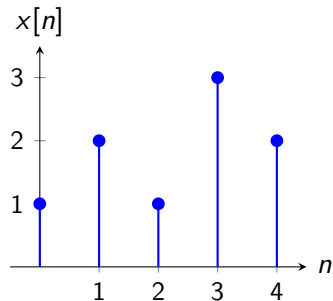
- Takes values from a **finite set**
- Often equally spaced
- Example: digital levels (0,1,2,...)

Visual Comparison

Continuous-Valued



Discrete-Valued

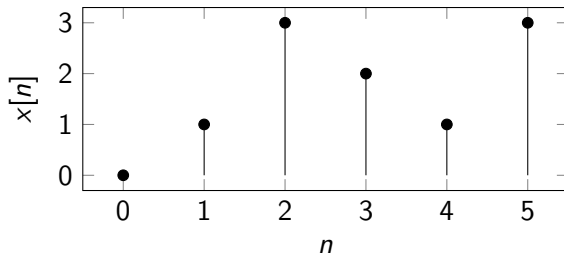


Digital Signals

Definition:

- Discrete in time AND discrete in value

Example: Signal values limited to $\{0,1,2,3\}$



Deterministic Signals

Definition:

- Fully described by a mathematical expression or rule

Properties:

- No uncertainty
- Future values are predictable

Example:

$$x(t) = \sin(t)$$

Random Signals

Definition:

- Cannot be precisely predicted
- No exact mathematical expression

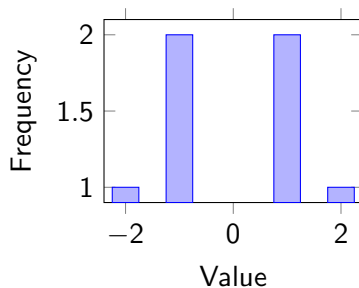
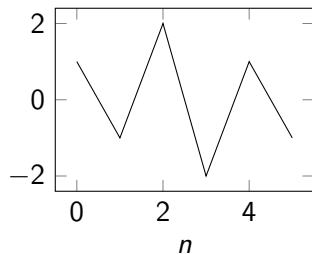
Examples:

- Noise
- Speech signal
- Seismic signals

Key Idea:

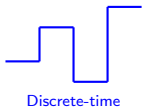
- Use **probability and statistics** for analysis

Random Signal Example



Insight: Different signals may look different but share similar statistical properties.

Discrete-Time vs. Discrete-Valued Signals



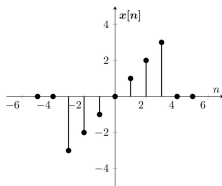
Discrete-Time vs Discrete-Valued Signals

Discrete-Time

- Time index (horizontal axis)
- Defined only at:

$$n = 0, 1, 2, 3, \dots$$

- Example: sampled audio
- ⇒ Amplitude can still be continuous (e.g., 2.3, -1.78)

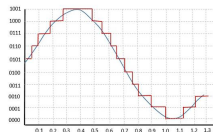


Discrete-Valued

- Amplitude (vertical axis)
- Takes fixed values:

$$x[n] \in \{0, 1, 2, 3\}$$

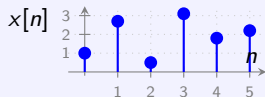
- Due to quantization
- ⇒ Can exist at all times



Discrete-Time vs. Discrete-Valued — The Key Difference

Discrete-Time Signal (continuous-valued)

Definition: Exists *only* at integer indices n ; amplitude $x[n]$ may be *any* real number.

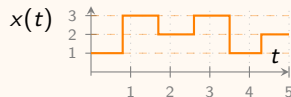


$n \in \mathbb{Z}$ (integer index) $x[n] \in \mathbb{R}$ (any real amplitude)

Example: Audio at 44 100 Hz; $t = n/f_s$. Amplitude can be 0.731, -0.223 , etc.

Discrete-Valued Signal (continuous-time)

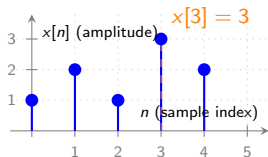
Definition: Amplitude restricted to finite levels $\{L_1, \dots, L_N\}$; time t remains *continuous*.



$t \in \mathbb{R}$ (continuous time) $x(t) \in \{L_1, \dots, L_N\}$
(finite levels)

Example: Thermostat reporting only $\{18, 19, 20, 21, 22\}^{\circ}\text{C}$ — reads continuously but outputs five levels.

Reading the Stem Plot — What Are n and $x[n]$?



Sample at $n = 3$: stem height = $x[3] = 3$
Real time: $t = 3 \times T_s$ seconds

Horizontal axis — n

n is the **sample index**: integer counter $n \in \mathbb{Z}$. Dimensionless on its own; becomes physical time via:

$$t = n \cdot T_s = \frac{n}{f_s}$$

where f_s is the sampling frequency (Hz).
If $f_s = 8000$ Hz, then $n = 160 \Rightarrow t = 20$ ms.

Vertical axis — $x[n]$

$x[n]$ is the **amplitude** at sample n . Square brackets $[\cdot]$ denote discrete-time; round brackets (t) denote continuous-time. The *height* of each stem equals $x[n]$.

In a discrete-time signal $x[n] \in \mathbb{R}$. After quantisation it belongs to a finite set of levels.

“Discrete in Time **AND** Value” — Digital Signals

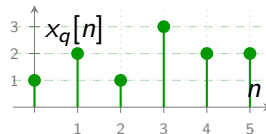
What does it mean?

A signal that is *simultaneously*:

- **Discrete in time:** only at integer indices n
- **Discrete in value:** amplitude rounded to one of 2^B levels by a B -bit ADC

Such a signal is called a **digital signal**.

Digital signal (discrete in time & value)



The ADC process

- 1 **Sample** at $f_s \Rightarrow$ discrete time
- 2 **Quantise** to B bits $\Rightarrow$ discrete value

$$x_q[n] = \Delta \cdot \left\lfloor \frac{x[n]}{\Delta} + \frac{1}{2} \right\rfloor, \quad \Delta = \frac{x_{\max} - x_{\min}}{2^B}$$

Property	Discrete-Time	Digital
Time n	$\mathbb{Z}$	$\mathbb{Z}$
Value $x[n]$	$\mathbb{R}$	$\{L_1 \dots L_{2^B}\}$
Storable	no	yes

Example: 16-bit audio at 44 100 Hz gives $T_s \approx 22.7 \mu\text{s}$ and $2^{16} = 65\,536$ amplitude levels.

Summary — Four Signal Classes

	Continuous Value	Discrete Value
Continuous Time	Analog signal $x(t) \in \mathbb{R}, t \in \mathbb{R}$ e.g. microphone voltage	Quantised analog $x(t) \in \{L_i\}, t \in \mathbb{R}$ e.g. thermostat output
Discrete Time	Sampled signal $x[n] \in \mathbb{R}, n \in \mathbb{Z}$ e.g. ideal DSP computation	Digital signal $x[n] \in \{L_i\}, n \in \mathbb{Z}$ e.g. MP3, JPEG, computers

Key takeaway: Discrete in time *and* value = **digital**. Only digital signals can be stored and processed by computers. Real-world signals need an **ADC** to become digital.

Mathematical Representation of Signals

Simple Forms

- Linear: $s(t) = 20t$
- Quadratic: $s(t) = 5t^2$
- Exponential: $s(t) = e^{-\alpha t}$

Multi-Variable

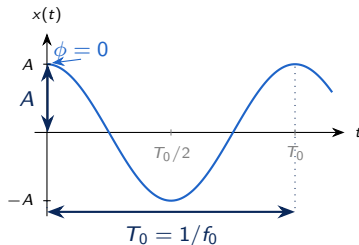
$$v(x, y) = 3x + 2xy + 10y^2$$

Pixel intensity at (x, y) .

The Sinusoid

$$x(t) = A \cos(2\pi f_0 t + \phi)$$

A amplitude
 f_0 frequency (Hz)
 ϕ phase (rad)
 $T_0 = 1/f_0$ period



Summary

- Continuous vs Discrete $\rightarrow$ based on values
- Digital signal $\rightarrow$ discrete time + discrete values
- Quantization introduces error
- Deterministic $\rightarrow$ predictable
- Random $\rightarrow$ unpredictable, analyzed statistically

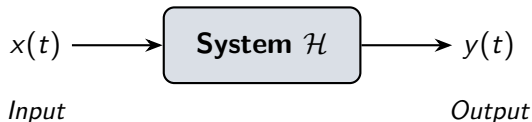
What is a System?

Definition

A **system** is any process or device that **transforms** an input signal into an output signal according to a defined rule or physical law.

$$y(t) = \mathcal{H}\{x(t)\}$$

where $\mathcal{H}$ denotes the system operator.



Real-World Systems

- Noise-cancelling headphones
- Image enhancement software
- Mobile radio receiver
- Heart rate monitor circuit

Common system examples:

- Amplifier: $y(t) = G \cdot x(t)$
- Integrator: $y(t) = \int_{-\infty}^t x(\tau) d\tau$
- Time delay: $y(t) = x(t - T_d)$

Signal Processing

Definition

Signal processing refers to operations performed on a signal to **extract, enhance, compress, or transmit** the information it contains.

Common operations:

- **Filtering** — remove unwanted frequency components
- **Amplification** — increase signal strength
- **Compression** — reduce data size (MP3, JPEG)
- **Modulation** — encode signal for transmission

Everyday Applications

- Removing wind noise during a phone call
- JPEG / HEIF compression of smartphone photos
- Active noise cancellation in earbuds
- ECG baseline wander removal in hospitals
- Radar target detection in aviation

Why It Matters

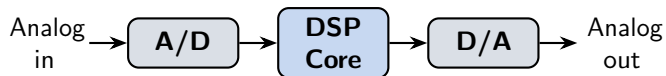
Without signal processing there would be no smartphones, no digital music, no MRI machines, and no internet video calls.

Digital Signal Processing (DSP)

Definition

DSP is the use of digital computers or microprocessors to process signals that have been converted from analog form into a sequence of binary numbers.

The core processing flow:



- **A/D Converter:** Samples & quantises the analog signal
- **DSP Core:** Executes algorithms (filtering, FFT, ...)
- **D/A Converter:** Reconstructs analog output if needed

Digital Signal Processing (DSP)

Example: A Mobile Phone Call

1. Voice (analog) enters the microphone
2. A/D: sampled at 8 kHz, 16-bit
3. Compressed using AMR codec
4. Transmitted as packets over 4G/5G
5. Decompressed at the receiver
6. D/A: converted back to sound

Analog vs. Digital: Detailed Comparison

Feature	Analog	Digital
Signal form	Continuous, smooth values	Discrete binary values (bits)
Noise immunity	Low — noise accumulates	High — error detection & correction
Storage	Degrades over time (tape, film)	Lossless (hard drives, cloud)
Flexibility	Fixed hardware design	Reprogrammable via software
Precision	Limited by component tolerances	Set exactly by bit depth
Speed	Very fast	Requires fast ADC clock

Neither is Universally Better

Analog excels in raw speed and simplicity. Digital excels in accuracy, flexibility, and programmability. Modern systems combine both.

Thank You